UNIVERSITÄT
LEIPZIG
Medizinische Fakultät

Architecture of IS in GW / Information management in clinical research

Transformation of a CSV data set to HL7 FHIR

Leipzig, date

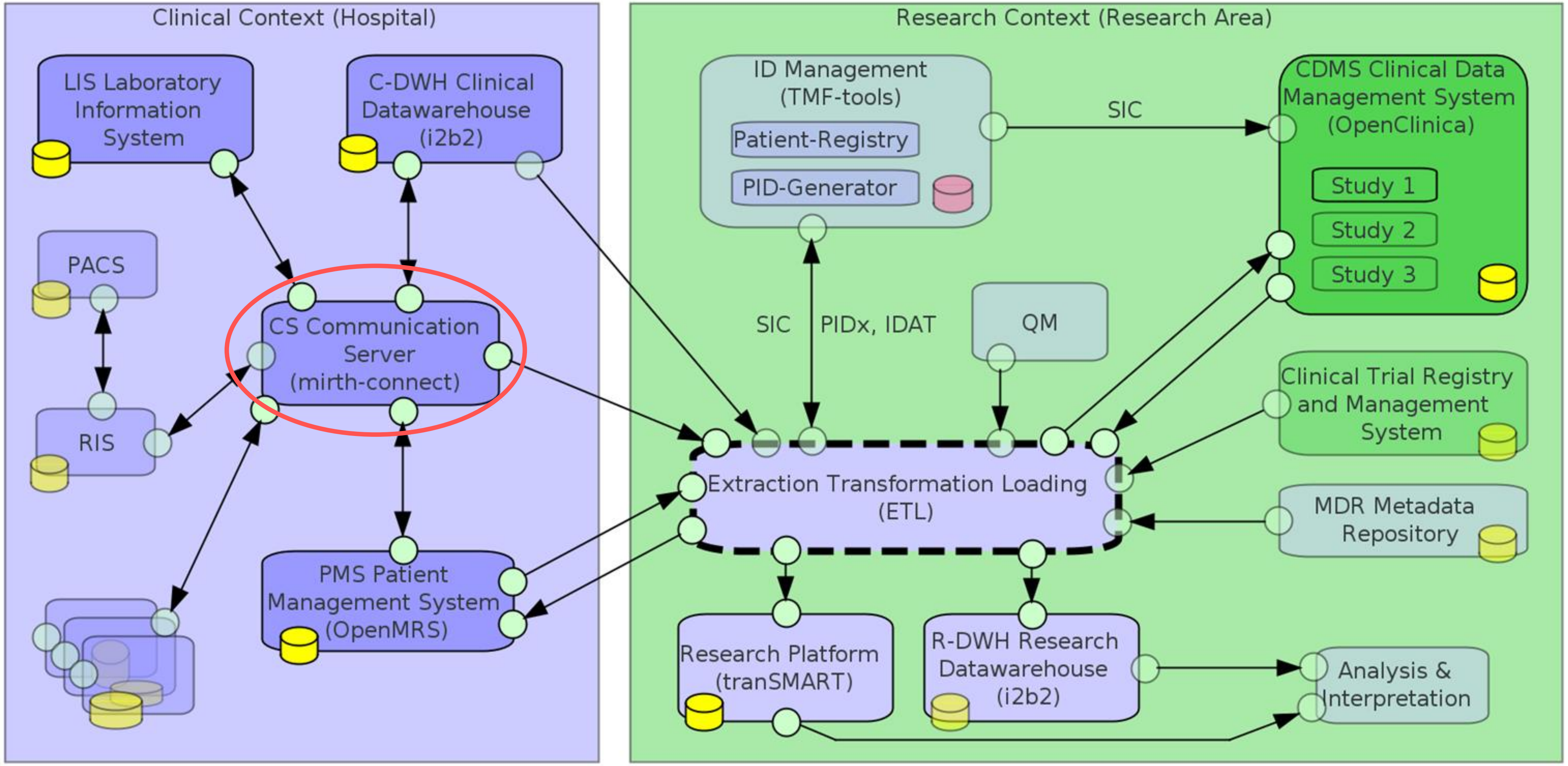
imise.

CONTENTS

COMMUNICATION SERVER

- **Function**
  - Central distribution of messages
  - Filter
  - Transformation
  - Avoidance of n*m communication interfaces
- **Software products**:
  - eGate
  - Kodix (UKL)
  - **NextGen Connect** (formerly Mirth Connect)

COMMUNICATION SERVER



FAST HEALTHCARE INTEROPERABILITY RESOURCES (FHIR)

- Consisting of consistent, easy-to-understand resources
- 80-20 principle: the standard (currently R4) covers most applications, with the rest covered by extensions
- Web technology (Restful API) + human readability
- Open source (specification, tools, servers, libraries)
- Maintained by HL7
- Used in Germany as part of the MII (core data set)

Patient	I		Patient
id	S Σ	0..1	string
meta	S Σ	0..1	Meta
identifier	S Σ	*0..**	*Identifier*
active	Σ ?!	0..1	boolean
name	S Σ	*0..**	*HumanName*
telecom	Σ I	0..*	ContactPoint
gender	S Σ	0..1	code Binding
birthDate	S Σ	0..1	date
deceased[x]	S Σ ?!	0..1	
address	S Σ	*0..**	*Address*
maritalStatus		0..1	CodeableConcept Binding
multipleBirth[x]		0..1	
photo	I	0..*	Attachment
contact	I	0..*	BackboneElement
communication		0..*	BackboneElement
generalPractitioner	I	0..*	Reference(Organization | Practitioner | Practit...
managingOrganization	Σ I	0..1	Reference(Organization)
link	S Σ ?!	0..*	BackboneElement

https://www.medizininformatik-initiative.de/Kerndatensatz/Modul_Person_Version_2/MIIIGModulPerson-TechnischeImplementierung-FHIR-Profile-PatientInPatient.html

PURPOSE OF THE EXERCISE

- Transformation from CSV to FHIR
  - Mapping to FHIR with MII-KDS
- Creating channels with Mirth-Connect
  - Receiving data
  - Transforming
  - Outputting data in FHIR format
- Sending FHIR data to a FHIR server
  - Verifying the transformed FHIR data with ClinFHIR

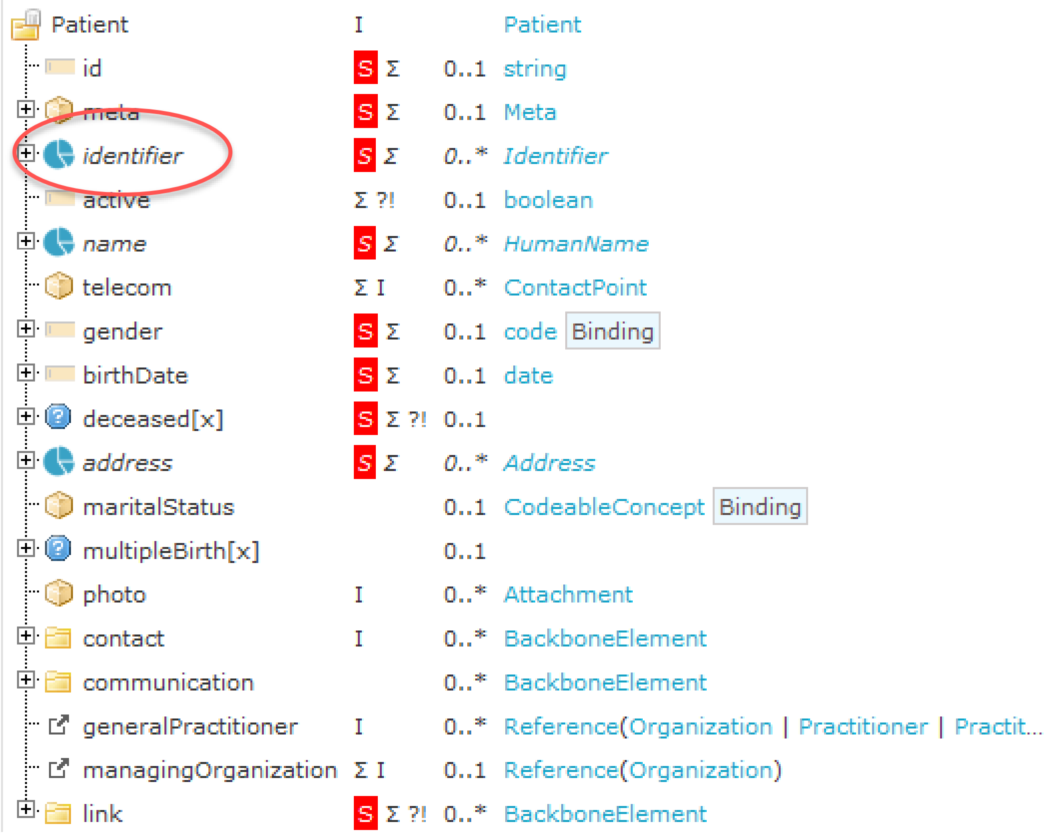
TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

PROCESS

1. Input data
   - CSV data set
2. Transformation with Mirth-Connect
   - Mapping of CSV columns to FHIR attributes
   - Based on FHIR message template
3. Output
   - JSON bundle in FHIR format
   - For sending to a FHIR server

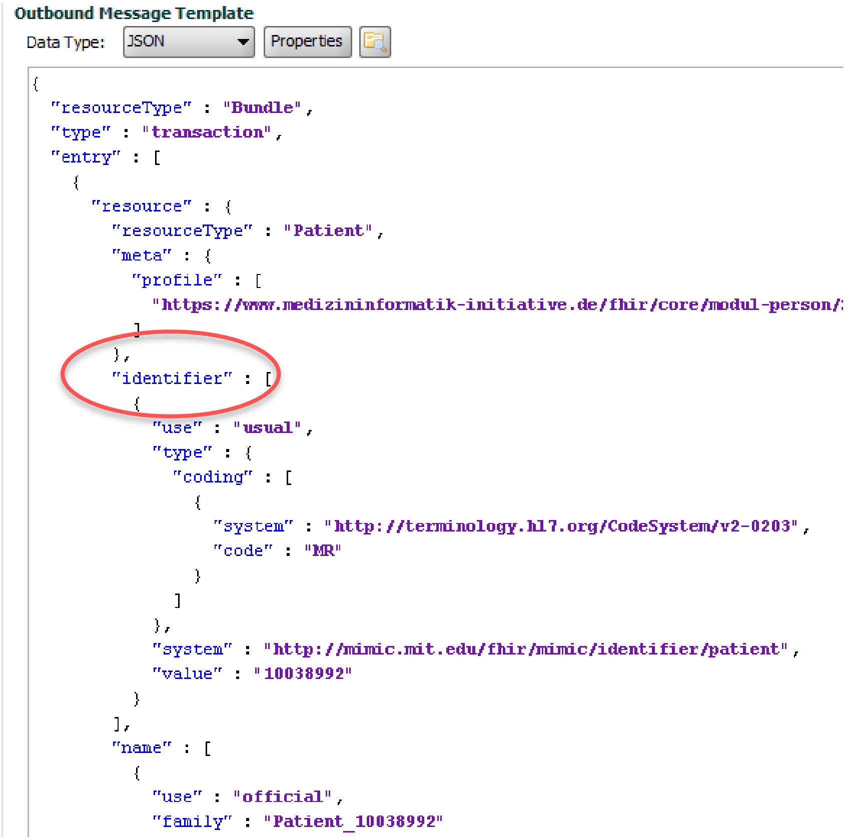
TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

FHIR TEMPLATE: MII PATIENT



https://www.medizininformatik-initiative.de/Kerndatensatz/Modul_Person_Version_2/MIIIGModulPerson-TechnischeImplementierung-FHIR-Profile-PatientInPatient.html

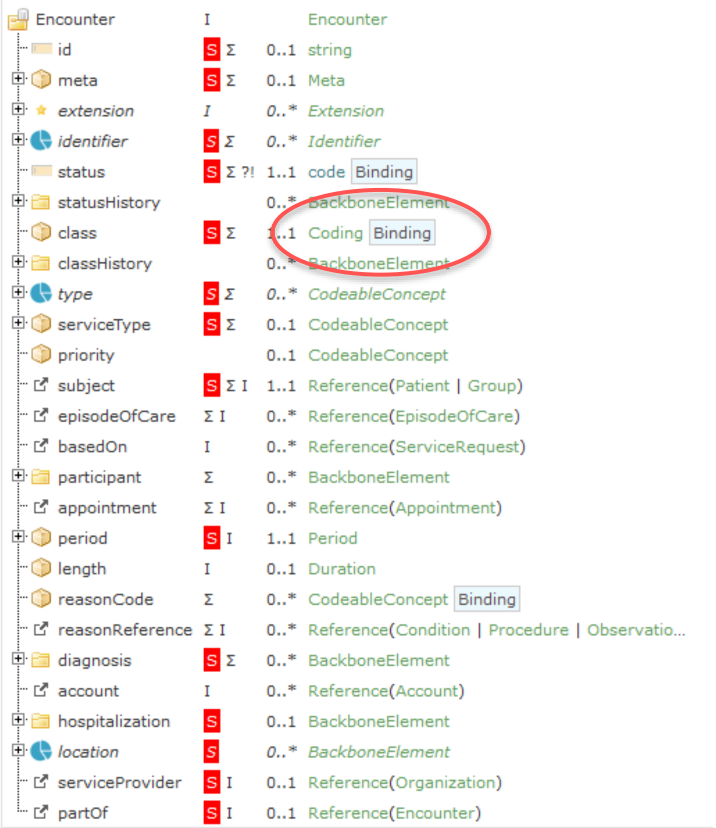
Template for FHIR messages to be sent



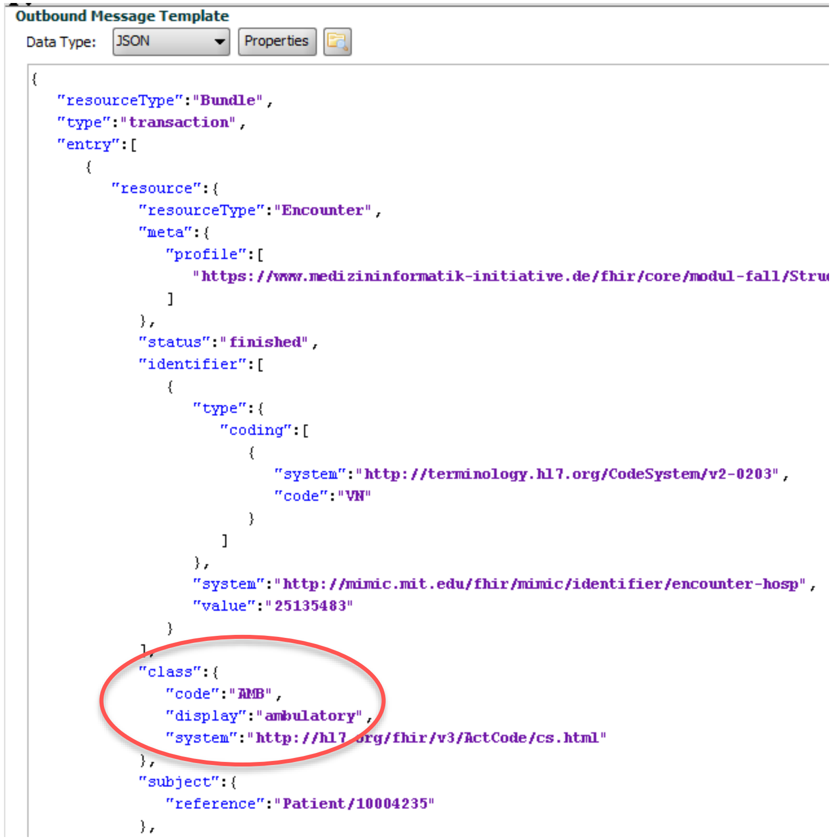
Mirth-Connect Patient Source Transformer

TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

FHIR TEMPLATE: MII ENCOUNTER



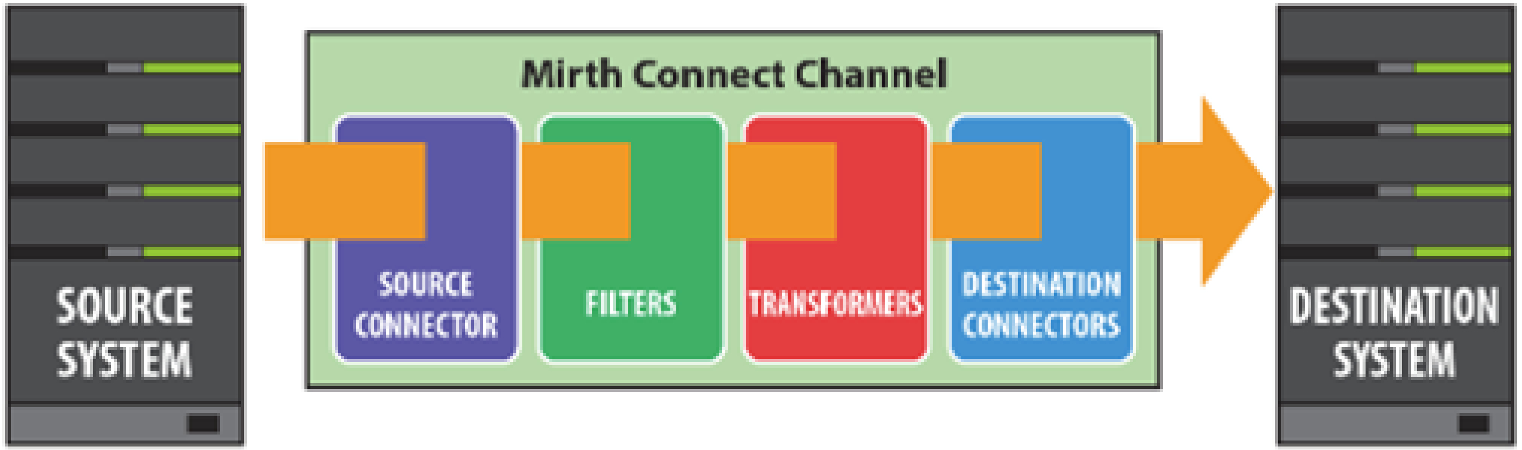
https://www.medizininformatik-initiative.de/Kerndatensatz/Modul_Fall_Version_2/MIIIGModulFall-TechnischeImplementierung-FHIRProfile-EncounterKontaktGesundheitseinrichtung.html



Code system: https://www.medizininformatik-initiative.de/Kerndatensatz/Modul_Fall_Version_2/MIIIGModulFall-TechnischeImplementierung-FHIRProfile-EncounterKontaktGesundheitseinrichtung.html

MIRTH-CONNECT

CHANNEL STRUCTURE



https://www.adldata.org/download/Manuals/Mirth_Connect_3.4_Users_Guide.pdf

1. **Source** (input)
   - Read CSV data set
   - ➢ File Reader (defined folder)
2. **Filters & Transformer** (processing)
   - Transforming input data CSV to FHIR template
   - Mapping CSV columns to FHIR attributes
3. **Destination** (output)
   - Send the transformed FHIR JSON messages to the FHIR server (REST API)
   - ➢ Http sender

MIRTH-CONNECT

2. SOURCE **TRANSFORMER**

- Transformation of the data set to FHIR
- Mapping CSV columns to FHIR attributes
- Transformer Step: attribute to be mapped

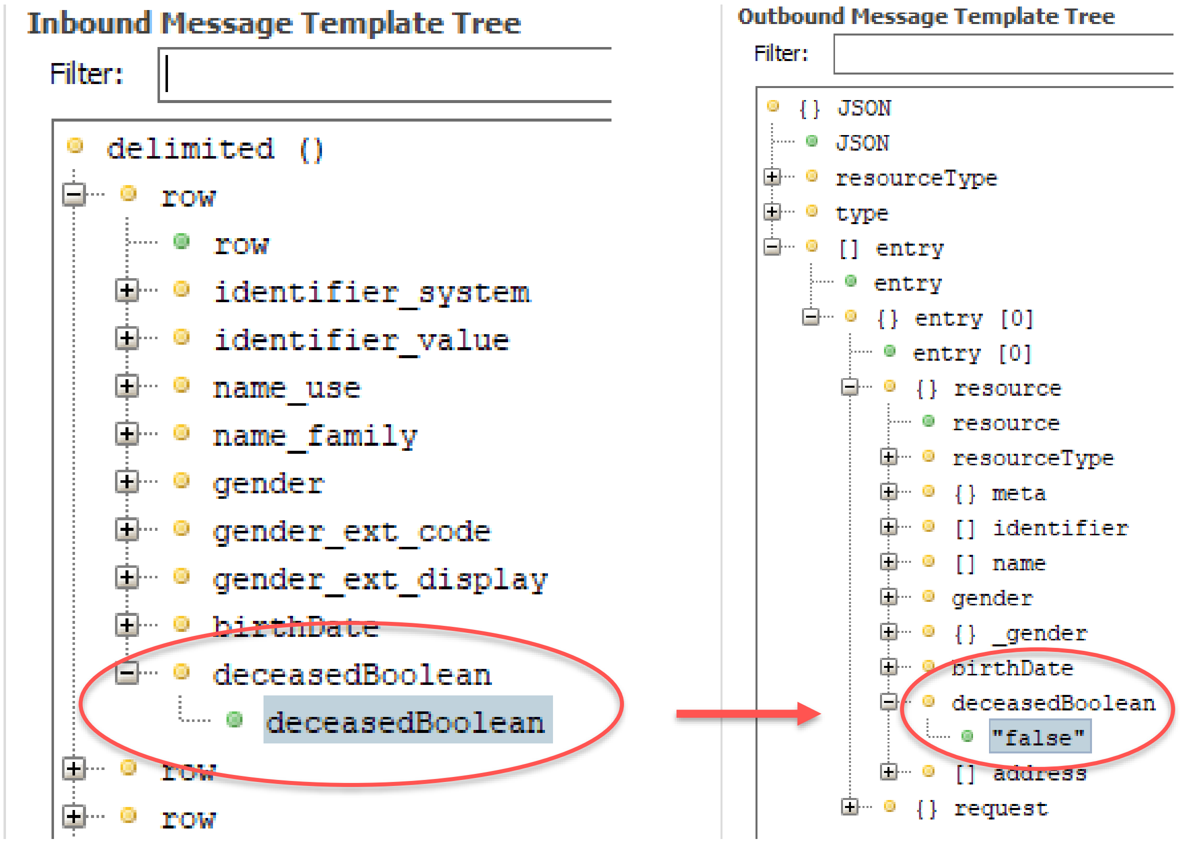
- **Input**: CSV file
- **Output**: FHIR message template

TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

MII PATIENT TRANSFORMATION: EXAMPLE

Input: CSV file

- All rows and columns as tree structure
- Mapping of the first row sufficient

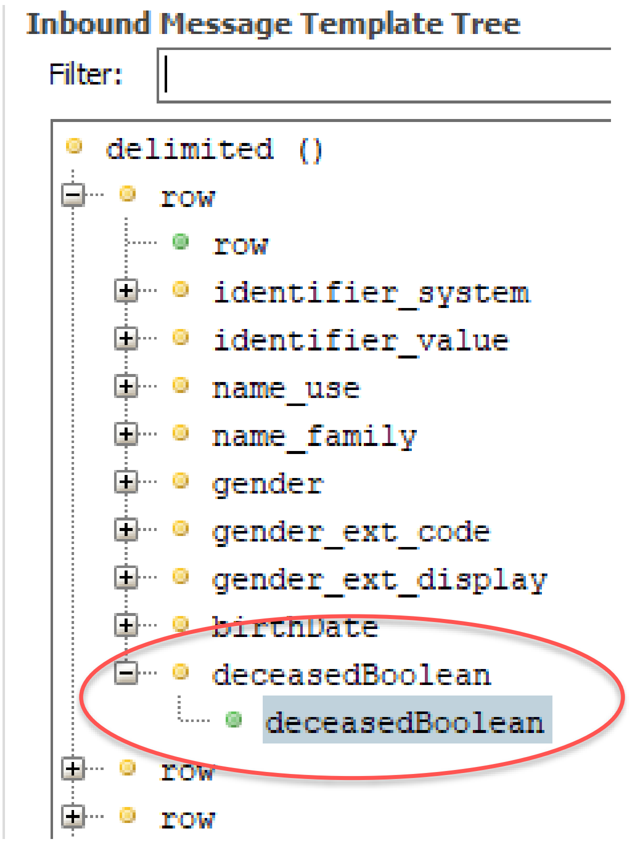


Output: FHIR JSON template

- A JSON tree

TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

MII PATIENT TRANSFORMATION: EXAMPLE



- Mirth-Connect Transformer
- **Input** CSV:

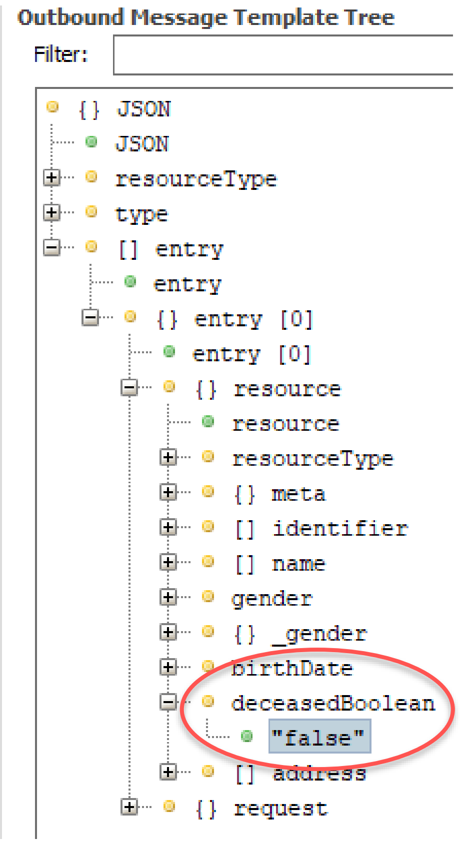
Mirth-Connect syntax

➢ msg['row'][0]['deceasedBoolean'].toString()

Target variable

TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

MII PATIENT TRANSFORMATION: EXAMPLE



- Mirth-Connect Transformer
- **Output** FHIR message template:

Mirth-Connect Syntax

➢ tmp['entry'][0]['resource']['deceasedBoolean']

Target variable

TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

TRANSFORMATION TERMS

- Transformer Step
  - 1 step for each attribute to be mapped
- Step Type: **Message Builder**
  - Replaces attribute in FHIR template with attribute from CSV data record

- **Message Segment** (Output)
  - Text field for attribute from FHIR template
- **Mapping** (Input)
  - Text field for attribute from CSV data record

TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

MII PATIENT TRANSFORMATION: EXAMPLE

Enabled	#	Name	Type
☑	0	identifier_system	Message Builder
☑	1	identifier_value	Message Builder
☑	2	name_use	Message Builder
☑	3	gender_code	Message Builder
☑	4	gender_display	Message Builder
☑	5	birthdate	Message Builder
☑	6	deceased	Message Builder

Step | Generated Script

Message Segment: tmp['entry'][0]['resource']['identifier'][0]['system']

Mapping: msg['row'][0]['identifier_system'].toString()

TASKS (SEE EXERCISE SHEET)

1. Mirth Connect
   - **1.1 - 1.2** Create channel
   - **1.3.1** Source connector
   - **1.3.2** Destination connector
   - **1.4 - 1.5** Extend transformer
   - **1.6** Deploy channels and send FHIR messages

1. ClinFHIR
   - **2.1** View transformed patients and cases
   - **2.2 - 2.3** Answer questions

TASKS (SEE EXERCISE SHEET)

Exercise sheet in **GitHub**:

https://github.com/IMISE/MI-Lab-E02-CSV-to-FHIR

➔ Under /Exercise/Exercise_sheet.pdf

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Medizinische Fakultät

THANK YOU!

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